HW2

Monday, 16 January 2023 4:17 PM

Aq phase: $Q^+X^- + A^- \underset{k_2}{\overset{k_1}{\rightleftharpoons}} Q^+A^- + X^-$

Organic phase: $Q^+X^- + AB \overset{k_3}{\longleftarrow} Q^+A^- + BX$

Catalyst: QX
Complex: QA
Products: AB, X^-

Overall: $A^- + BX \longrightarrow AB + X^-$

$\frac{k_2}{k_1} = K \equiv$ equilibrium constant $= \frac{[QA]_a[X^-]}{[QX]_a[A^-]}$; Well-stirred reactor: uniform conc. of all species can be assumed where interphase conc. is equal to the bulk conc.

$[QA]_a = [QA]_{a,e}$; $[QX]_a = [QX]_{a,e}$

$[QA]_a = [QA]_{o,e}$; $[QX]_o = [QX]_{o,e}$

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$X = (1-\varphi)\frac{[QA]_o}{Q_T}$

(i) [a] Show $\frac{dU}{d\tau} = -UV$

In organic phase (for constant volume): $\frac{dc_j}{dt} = \alpha_j r$; $QA + BX \xrightarrow{k_3} QX + AB$

Assuming elementary reaction: $r_o = -k_3[QA]_o[BX]$

$\therefore \frac{d[BX]}{dt} = -k_3[QA]_o[BX]$; $\alpha_j = 1$

$\div$ by $[BX]_i$

$\frac{1}{[BX]_i}\frac{d[BX]}{dt} = -k_3\frac{[BX]}{[BX]_i}[QA]_o$

$\times \frac{\varphi}{Q_T}$

$\frac{\varphi}{Q_T\cdot[BX]_i}\frac{d[BX]}{dt} = -k_3\frac{[BX]}{[BX]_i}[QA]_o\cdot\varphi$

$\frac{d\left(\frac{[BX]}{[BX]_i}\right)}{d\left(\frac{t\cdot k_3\cdot Q_T}{\varphi}\right)} = -\frac{[BX]}{[BX]_i}\frac{\varphi[QA]_o}{Q_T}$

$$\boxed{\frac{dU}{d\tau} = -U\cdot V}$$

[b] Show: $\frac{dV}{d\tau} = -P_1(V - P_3X) - P_2UV$; $P_1 \equiv \frac{k_L a}{k_3 Q_T}$; $P_2 \equiv \varphi\frac{[BX]_i}{Q_T}$; $P_3 = \pi_{QA}\frac{\varphi}{1-\varphi} = \frac{[QA]_{o,e}}{[QA]_{a,e}}$

$$\frac{d[QA]_o}{dt} = -k_3[QA]_o[BX] + K_L A([QA]_{o,e} - [QA]_o) \quad ; \quad \varphi = \frac{V_o}{V_T} \text{ (dispersed phase fraction)} \quad ; \quad V_T = V_o + V_a$$

$\frac{d[QA]_o}{dt}$: change in conc. of $[QA]_o$ — $-k_3[QA]_o[BX]$: Reaction in organic layer — $K_L A([QA]_{o,e} - [QA]_o)$: transport of $[QA]_o$ from aq. layer

; A = interfacial area (effective mass transfer area)

$\left(\times \frac{\varphi}{k_3 \cdot Q_T}\right)$

$$\frac{d}{dt}\left(\frac{\varphi}{Q_T}[QA]_o\right) = -k_3[QA]_o[BX]\frac{\varphi}{Q_T} + \frac{\varphi}{Q_T}\frac{K_L A}{k_3}([QA]_{o,e} - [QA]_o)$$

$$\Rightarrow \frac{1}{k_3}\frac{dV}{dt} = -[QA]_o\frac{[BX]}{[BX]_i}\cdot\left([BX]_i\frac{\varphi}{Q_T}\right) + \frac{K_L(\varphi A)}{k_3 Q_T}([QA]_{o,e} - [QA]_o) \quad ; \quad \varphi A = a$$

$\left(\times \frac{\varphi}{Q_T}\right)$

$$\Rightarrow \frac{1}{\frac{k_3 Q_T}{\varphi}}\frac{dV}{dt} = -U\cdot P_2\left([QA]_o\frac{\varphi}{Q_T}\right) + P_1\left(\frac{\varphi}{Q_T}[QA]_{o,e} - \frac{\varphi}{Q_T}[QA]_o\right)$$

$$\frac{dV}{d\tau} = -P_2 UV + P_1\left(\frac{\varphi}{Q_T}\frac{[QA]_{o,e}}{[QA]_{a,e}}\cdot[QA]_a - V\right) \quad ; \quad [QA]_{a,e} = [QA]_a$$

$$\frac{dV}{d\tau} = -P_2 UV + P_1\left(\frac{\varphi}{(1-\varphi)}\pi_{QA}\cdot\frac{(1-\varphi)}{Q_T}[QA]_a - V\right)$$

$$\frac{dV}{d\tau} = -P_2 UV + P_1\left(P_3\cdot X - V\right)$$

$$\therefore \boxed{\frac{dV}{d\tau} = -P_2 UV - P_1(V - P_3 X)}$$

[c] Show: $\frac{dW}{d\tau} = -P_1(W - P_4 Y) + P_2 UV$; $P_4 = \pi_{QX}\frac{\varphi}{1-\varphi}$

Similar to previous part: $\frac{d[QX]_o}{dt} = k_3[QA]_o[BX] - K_L A([QX]_o - [QX]_{o,e})$

$\frac{d[QX]_o}{dt}$: Rate of change of $[QX]$ — $k_3[QA]_o[BX]$: rate at which produced in organic phase — $K_L A([QX]_o - [QX]_{o,e})$: transport of $[QX]$ to aq. layer

$\left(\times \frac{\varphi^2}{k_3 Q_T^2}\right)$:

$$\frac{d\left(\frac{[QX]_o\varphi}{Q_T}\right)}{d\left(\frac{t\cdot k_3\cdot Q_T}{\varphi}\right)} = \varphi\frac{[BX]_i}{Q_T}\cdot\frac{[BX]}{[BX]_i}\cdot\frac{[QA]_o\varphi}{Q_T} - \frac{K_L(A\cdot\varphi)}{Q_T k_3}\left(\frac{[QX]_o\varphi}{Q_T} - \frac{[QX]_{o,e}}{[QX]_{a,e}}\cdot[QX]_a\cdot\frac{\varphi}{Q_T}\cdot\frac{\varphi}{(1-\varphi)}(1-\varphi)\right)$$

$$\Rightarrow \boxed{\frac{dW}{d\tau} = +P_2 UV - \frac{k_L a}{Q_T k_3}\left(W - P_4 Y\right)}$$

[d] Show: $1 = V + W + X + Y$

Mole balance on catalyst (Q^+) ; $C_j = \frac{N_j}{V}$; $V_T = V_o + V_a$

$$V_a \cdot [QX]_a + [QA]_a V_a + [QA]_o V_o + [QX]_o \cdot V_o = Q_T \cdot V_T$$

$$\Rightarrow \frac{V_a}{V_T}\frac{[QX]_a}{Q_T} + \frac{V_a}{V_T}\frac{[QA]_a}{Q_T} + \frac{V_o}{V_T}\frac{[QA]_o}{Q_T} + \frac{V_o}{V_T}\frac{[QX]_o}{Q_T} = 1$$

$$\Rightarrow \frac{(V_T - V_o)}{V_T}\frac{[QX]_a}{Q_T} + \frac{(V_T - V_o)}{V_T}\frac{[QA]_a}{Q_T} + \Phi\frac{[QA]_o}{Q_T} + \Phi\frac{[QX]_o}{Q_T} = 1$$

$$\Rightarrow (1-\Phi)\frac{[QX]_a}{Q_T} + (1-\Phi)\frac{[QA]_a}{Q_T} + V + W = 1$$

$$\therefore \boxed{Y + X + V + W = 1}$$

(e) Show $P_5(1-Z) = P_2(1-U) + V + X$; $Z = \frac{[A^-]}{[A^-]_i}$; $P_5 = (1-\Phi)\frac{[A^-]_i}{Q_T}$

Overall reaction: $A^- + BX \rightarrow AB + X^-$

Mole balance on A^- species:

$$\Delta N_{A^-} = N_{QA,a} + N_{QA,o} + N_{AB}$$

Converting to concentrations, ;

(Starting w/ no initial AB)

$$\xi = \frac{\cancel{[AB]_i}^{\,0} - [AB]}{1} = \frac{[BX]_i - [BX]}{-1}$$

$$\xi = [AB] = [BX]_i - [BX]$$

$$V_a([A]_i - [A]) = V_a[QA]_a + V_o[QA]_o + [AB]\cdot V_o$$

$$V_a([A]_i - [A]) = V_a[QA]_a + V_o[QA]_o + ([BX]_i - [BX])V_o$$

Dividing by total volume and Q_T:

$$\Rightarrow \frac{V_a}{V_T \cdot Q_T}([A]_i - [A]) = \frac{V_a}{V_T Q_T}[QA]_a + \frac{V_o}{V_T Q_T}[QA]_o + \frac{V_o}{V_T Q_T}([BX]_i - [BX])$$

$$\Rightarrow (1-\Phi)\frac{[A]_i}{Q_T}\left(1 - \frac{[A]}{[A]_i}\right) = \frac{[BX]_i}{Q_T} - \Phi\cdot\left(1 - \frac{[BX]}{[BX]_i}\right) + \frac{(1-\Phi)}{Q_T}[QA]_a + \frac{\Phi}{Q_T}[QA]_o$$

$$\Rightarrow \boxed{P_5(1-Z) = P_2(1-U) + X + V}$$

(f) Show: $KP_5YZ = X[P_2(1-U) + 1 - W - Y]$

Mole balance on $[X^-]$: ; $K = \frac{[QA]_a[X^-]}{[A^-][QX]_a} = \frac{k_1}{k_2}$; $\xi = \frac{[X^-]_i - [X^-]}{1} = \frac{[A^-]_i - [A^-]}{-1}$

$$N_{X^-,i} = N_{X^-,f}$$

$[BX]_i \cdot V_0 + [X^-]_i V_a + [QX]_i \cdot V_a = V_0 \cdot [BX] + V_0 [QX]_0 + V_a \cdot [X^-] + V_a \cdot [QX]_a$; Assuming all initial $[QX]_i = [QX]_{a,i} = Q_T$

Dividing by V_T:

$\Rightarrow (1-\Psi)([X]_i^{\,0} - [X^-]) + \Psi([BX]_i - [BX]) = \Psi[QX]_0 + (1-\Psi)([QX]_a - Q_T)$

$\Rightarrow -(1-\Psi)[X^-] + \Psi[BX]_i\left(1 - \frac{[BX]}{[BX]_i}\right) = \Psi[QX]_0 - (1-\Psi)Q_T + (1-\Psi)[QX]_a$

$\times \frac{(1-\Psi)[QA]_a}{Q_T^2}$

$$\Rightarrow \frac{(1-\Psi)^2[QA]_a[X^-]}{Q_T^2} = -\frac{\Psi(1-\Psi)[QX]_0[QA]_a}{Q_T^2} - \frac{(1-\Psi)^2[QA]_a[QX]_a}{Q_T^2} + \frac{(1-\Psi)[QA]_a}{Q_T} + \frac{(1-\Psi)\Psi[QA]_a[BX]_i}{Q_T^2}\left(1 - \frac{[BX]}{[BX]_i}\right)$$

$$= \frac{(1-\Psi)[QA]_a}{Q_T}\left[1 + \frac{\Psi}{Q_T}[BX^-](1-U) - \frac{\Psi[QX]_0}{Q_T} - \frac{(1-\Psi)[QX]_a}{Q_T}\right]$$

$$\Rightarrow K\frac{(1-\Psi)[A^-]_i}{Q_T}\frac{(1-\Psi)[QX]_a}{Q_T}\frac{[A^-]}{[A^-]_i} = \frac{(1-\Psi)[QA]_a}{Q_T}\left[\frac{\Psi[BX]_i}{Q_T}\left(1 - \frac{[BX]}{[BX]_i}\right) + 1 - \frac{\Psi[QX]_0}{Q_T} - \frac{(1-\Psi)[QX]_a}{Q_T}\right)$$

$$\therefore \boxed{KP_5YZ = X[P_2(1-U) + 1 - W - Y]}$$

Rest of HW completed on Python (see attached)